Hackathon Challenge

Apply the Full ML Pipeline to a New Clinical Dataset

QLS–MiCM Workshop

The Goal

Build the **best-performing classifier** for predicting *diabetes risk* on a new clinical dataset.

You will:

- use everything learned so far
- explore models *creatively*
- compare performance fairly
- explain your reasoning clearly

What You're Given

- A dataset: **clinical risk**
- A **baseline pipeline** with preprocessing + logistic regression
- A train/test split
- Code that computes performance metrics

This is your starting point — nothing more.

What You Can Change (Important)

You may modify **only the model step** inside the existing pipeline.

You may:

- swap in any scikit-learn classifier
- adjust its hyperparameters
- run multiple experiments
- document your choices

You may **NOT**:

- change preprocessing
- change the train/test split
- alter or rebalance the data
- engineer new features
- manually inspect or label data
- use external libraries or pretrained models

All teams compete under the **same constraints**

How We Will Evaluate

Each team will present:

- **Model used**
- **Final test-set score (ROC AUC)**
- **Short explanation** of what you tried and why

Ranking is based on:

1. **ROC AUC**
2. Clarity of reasoning
3. Efficient exploration (not random guessing)

Team Workflow

1. Read the baseline output
2. Try different models
3. Tune them within reason
4. Record what you tried and why
5. Keep your notes short and clear
6. When time is up → report your best score

Tips (Not About Models)

- Communicate with your team
- Try things systematically
- Keep track of your experiments
- Don’t get stuck on one idea
- When time is almost up, stop changing and start summarizing

Good Luck!

May the best model win.